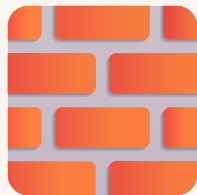


WEEK 2



Placing Blocks

Make the agent leave a trail

에이전트가 블록을 놓게 하기

BIG PICTURE

From walking to building 🏗️



Last week the agent walked. This week it builds.

걷기에서 짓기로

HOW TO COUNT

Count the axis — one block, one number



1 Stand on the start block — that spot is $(0, 0, 0)$.

2 $x \rightarrow$ walk **right**. Add **+1** for every block.

3 $y \rightarrow$ jump **up**. Add **+1** for every block.

4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

The place block command

The agent drops a block where you tell it: **down**, forward, or to a side.

```
place block down
```

This puts one block **under** the agent.

place block down = 발밑에 블록 놓기

CONCEPT

Move + place = a trail

place block down
drop a block

move forward
step ahead

move up by 1
go up 1

repeat ♻️
do it again

놓고 → 이동 → 놓고 → 이동

CONCEPT

Line vs stack



place + move forward
blocks in a line



place + move up
blocks stacked tall

앞으로 가면 한 줄, 위로 가면 탑

CODE

Blocks: a 3-block line

```
# place, step, place, step, place  
place block down  
move forward  
place block down  
move forward  
place block down
```

3 blocks in a straight line on the ground.

CODE

The same idea in Python

```
agent.place(DOWN)
agent.move(FORWARD, 1)
agent.place(DOWN)
agent.move(FORWARD, 1)
agent.place(DOWN)
```

DOWN = where the block goes (below the agent).

블록과 똑같은 동작이에요

PREDICT 🤖

Stacked, line, or floating?

Predict first — then click reveal

```
move up by 1  
place block down  
move up by 1  
place block down
```

Reveal output 🔍 reveal

VOCAB

New words today

place put a block where the agent is

block one cube in the Minecraft world

down the direction below the agent

stack blocks piled on top of each other

놓다

블록

아래로

쌓기



DEBUG

What's wrong with this code?

Goal: place a block, move forward, place another.

buggy

```
move forward  
place block down  
move forward
```

Think first — then click next.



DEBUG

The bug

Bug: only **one** block gets placed, and it starts with a wasted step. The order is wrong — place first, then move, then place again.

fixed

```
place block down  
move forward  
place block down
```

YOUR TURN!

Place 5 blocks

✓ Open your homework world + Code Builder

✓ Place **5 blocks** in different spots

✓ Use at least **2 block types** (stone, dirt, wood)

✓ Mix in `move forward` and `move up`

✓ Send a photo of what you placed



QUIZ

QUESTION 1 OF 3

What does `place block down` do?

Destroys a block.

Moves the agent down.

Puts one block below the agent.

Builds a whole wall.



QUIZ

QUESTION 2 OF 3

To make a line of blocks, you repeat:

place block down, then place block down.

place block down, then move forward.

move up, then move up.

turn right, then turn left.



QUIZ

QUESTION 3 OF 3

Which pair builds blocks going **upward**?

place block down, then move forward.

move down, then move down.

place block down, then move up by 1.

turn right, then move forward.

QUIZ

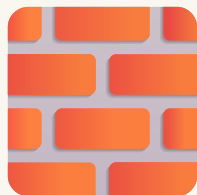
Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Blocks placed!

Next: loops to build a bridge.

다음 주: 반복문으로 다리 만들기